

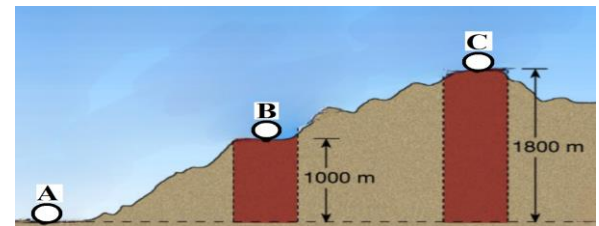
Engineering Physics II	Faculty of Engineering	Date: 3 /8/2025
B₁ EC/E	First Test - Modeule A	
Name:	Reg. No.:	

Section I: Answer all the following by putting true (T) or false (F): **6 Marks**

1) The electric flux decreases with area increase		
2) In Pascal principle, the force does not change.		
3) A suck pressure is positive gauge pressure		

Section II: Answer all the following by selecting the correct choice: **14 Marks**

1) Fully immersed object with density less than the liquid's density. The object will:				
a) Move up	b) Float	c) Sinking	d) be at rest	e) None of these
2) Two spheres come into contact and then separate. The charge on sphere B is:				
			A 3 C B 0 C	
a) 1.5 C	b) 0 C	c) - 1 C	d) 3 C	e) None of these
3) The force 8 N. If a distance changes from 2 m into 8 m. The force is:				
a) 1 N	b) 0.5 N	c) 16 N	d) 128 N	e) None of these
4) From the figure shown below. The pressures				



a) $P_A < P_B$	b) $P_C > P_B$	c) $P_C = P_A$	d) $P_A > P_C$	e) None of these
5) 25 N object displaces 25 N of water. The object is:				
a) Floating	b) Jumping	c) Flying	d) Sinking	e) None of these
6) Which of the following has positive charge:				
a) Electron	b) Neutron	c) Proton	d) Electron & Proton	e) None of these
7) Object with $4\text{ m} \times 3\text{ m} \times 2\text{ m}$ having a mass of $0.1x\text{ kg}$. The pressure in Pa on the ground is:				
a) $20x$	b) $0.83x$	c) $2x$	d) $0.083x$	e) None of these

Section III: Answer all the following by selecting the correct choice

Full and clear steps are required

20 Marks

Constant		
Electron: $m = 9.11 \times 10^{-31}\text{ kg}$	$K_e = 9 \times 10^9\text{ N.m}^2/\text{C}^2$	Gravity: $g = 10\text{ m/s}^2$
Proton: $m = 1.67 \times 10^{-27}\text{ kg}$	$G = 6.67 \times 10^{-11}\text{ N.m}^2/\text{kg}^2$	Water: $\rho_f = 1000\text{ kg/m}^3$

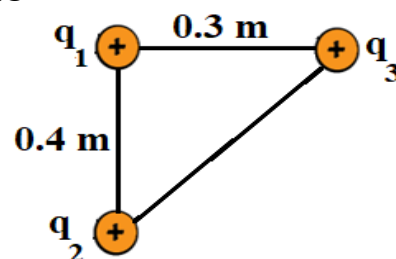
1) Two objects having $q_1 = 2\text{ }\mu\text{C}$ and $q_2 = 1\text{ }\mu\text{C}$, separated by 30 cm. The electrostatic force is:

a) $2.22 \times 10^{-11}\text{ N}$	b) 0.4 N	c) 0.2 N	d) 0.06 N	e) None of these
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2) A room has dimensions of 3 m and 3 m and a height of 4 m. If the air density is 1.21 kg/m^3 . The atmosphere force on the floor of the room is: Take $P_{\text{atm}} = 10^5 \text{ Pa}$

a) 900000 N	b) 600000 N	c) 800000 N	d) 500000	e) None of these
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3) The total electric force on q_3 is: $q_1 = 1 \text{ } \mu\text{C}$, $q_2 = 2 \text{ } \mu\text{C}$ and $q_3 = 3 \text{ } \mu\text{C}$



a) 0.46 N	b) 0.79 N	c) 0.37 N	d) 0.57 N	e) None of these
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4) Two opposite charges one is 4 times the other separated by distance r , if a ratio of charge to a distance r is $2 \text{ } \mu\text{C/m}$. The electric force on $3q$ placed at distance r from a point where $E = 0$ is:

a) 0 N	b) 0.36 N	c) 0.75 N	d) 1.33 N	e) None of these
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5) 500 N object in air and 150 N when it is fully immersed in a fluid of $\rho_f = 1500 \text{ kg/m}^3$. The volume of the metal is:

a) 0.01 m^3	b) 0.033 m^3	c) 0.1 m^3	d) 0.023 m^3	e) None of these
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